



The Right to Repair, the Right to Tinker, and the Right to Innovate

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Abstract: Creating new products by incorporating new and original ideas derived from learning the internal mechanisms and structures of machines and other objects at hand through the process of repairing or tinkering with them is fundamental to the innovation, which is a staple of human existence. Recently, however, increasing product complexity, technical constraints, and regulations have gradually narrowed the scope of the user's ability to tinker. This aspect has given momentum to the movement to explicitly reclaim the Right to Repair and the Right to Tinker. This paper thus outlines the process that led to recognition of the importance of these rights.

Keywords: open innovation, Right to Repair, Right to Tinker

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Introduction

The human species has been characterized since its origins by the act of creating new innovations based on ingenious improvements through insights obtained from studying objects, natural and man-made, that people (legitimately) possess by disassembling and/or experimenting on them. This aspect has been the driving force behind the progress of the human society.

Until recently, this aspect was such an obvious facet of everyday life that it hardly merited special attention. However, advances in computing and the Internet and their widespread use have dramatically reduced the cost of copying and distributing digital data, leading to illegal sharing of data online, resulting in prominent cases of harm to the interests of rights-holders. In response, the rights-holders designed technology-based protective measures in the form of copy control and access control, but this aspect quickly degenerated into a game of cat and mouse as technologies to work around such protection proliferated. A movement emerged to place legal restrictions on the very act of evading protection, and such “anti-circumvention of technological protection measures” can now be found in the intellectual property laws of many countries, including Japan.¹

These statutory restrictions, however, unwittingly impinged greatly on the freedom for anyone to tinker with things, which had historically been the underpinning of innovation. The impact is particularly dire in an era like this one wherein hardware and software are intimately related and the role of the software is relatively greater. For example, to tinker with or repair a smartphone, one needs to disassemble the smartphone, extract the firmware that controls it, research and modify it, and then return the modified

¹ See Table 1, [1][5]

Table 1. Major events of “the Right to Repair”

Year	Events	Notes
1956	The IBM Consent Decree	[1]
1998	Digital Millennium Copyright Act Section 117 & 1201	Anti-circumvention was introduced [2]
2001	Motor Vehicles Right To Repair Act S.2617 (failed)	For automobiles, the first of many attempts to establish the Right to Repair
2008	The U.S. Supreme Court favors a class action lawsuit challenging the carriers’ policies against unlocking phones	For mobile phones [3]
2012	Automotive Right to Repair passed in Massachusetts	The first in the U.S. [4]
2014	Digital Right to Repair Bill passed in South Dakota	The first successful bid of electronic Right to Repair [5]
2015	Library of Congress ruled in favor of repair-related exemption in DMCA	[6]
2019	20 states begin to consider the Right to Repair Bill	California, Georgia, Hawaii, Illinois, Indiana, Massachusetts, Minnesota, Missouri, Montana, North Dakota, Nevada, New Hampshire, New Jersey, New York, Oregon, South Dakota, Vermont, Virginia, Washington, West Virginia [7]

Source: U.S. PIRG (<https://uspig.org/feature/usp/right-repair>), repair.org, and various news outlets compiled by the author.

firmware to the phone; but being digital data, the firmware is a copyrighted work covered under the technological protective measures, making it illegal to freely copy or modify it in most cases; hence, circumventing or disabling the protective measures violates the law. In other words, one is not free to enhance a device even if one is its legitimate owner. Thus, a need to explicitly define and examine the previously self-evident Right to Tinker exists.

In this paper, we shall discuss the origins of the Right to Tinker and Right to Repair and their significance in the context of innovation.

The Brief History of “the Right to Repair”

The pre-history of the Right to Repair

In the period of domestic handicraft industry prior to the Industrial Revolution, it was a matter of course to repair one’s own home-made agricultural instruments/tools with no awareness of any particular “right” to do so. With the advent of mass production made possible by the Industrial Revolution, more complex production processes, and durable consumer goods, a clear distinction emerged between the producer and the consumer, thus raising the potential problem of the ability to repair. In his autobiography, automotive pioneer Henry Ford said of his automobiles, arguably the best single example of durable consumer goods, that “We want individuals who buy one of our products never to have to buy another,” and it was important that “You can take a 10-year-old car, and with parts that are available today, make it into a car of today with very little expense.” This aspect suggests that the parts must be interchangeable (Ford & Crowther, 1922, p. 124). For Ford, repairability was a crucial element of product design (Hanley, Kelloway, & Vaheesan, 2020).

The first time repairability as a consumer right was contested in a

court of law was the antitrust suit against IBM in 1956.² In a judgment handed down in a trial over whether its refusal to allow customers to repair machines they had leased contravened the law, IBM, which at the time held a dominant position in the market for mainframes, was mandated to offer parts and subassemblies to its customers at reasonable and nondiscriminatory prices.

“Planned obsolescence” as a counter concept

Ford’s notion of durable products was not exactly a paragon of economic rationality. With respect to encouraging new purchases by making repair parts for old models unavailable, Ford admitted that, “We have been told that this is good business, that it is clever business” (Ford & Crowther, 1922). Commencing in the 1920s, Ford Company organized dealers and repair shops into a network and provided repair parts only to certified shops and, in an attempt to evade competition on repair prices with independent outlets, made repairs impossible without using special tools supplied only to certified shops (McIntyre, 2000). At GM, led by Alfred Sloan, annual model changes were designed deliberately to create psychological obsolescence to stimulate replacement purchases by the consumers.

In addition to Sloan’s psychological obsolescence, other product development strategies designed to encourage replacement include functional obsolescence or adding new features to make the consumers feel that the previous models are old-fashioned and material obsolescence or intentionally shortening the product life (Yamamoto, 2012). Durable consumer goods such as cars and refrigerators, which had been assumed to have long lives thanks to repairability, found widespread popularity during the 1960s; then, in the 1970s, economics and management science came up with an aggressive approach to intentionally reduce durability by making

² United States v. IBM Corp., No. 72-344 (S.D.N.Y. 1956). See Table 1, [1].

repairs more difficult. The so-called Coase Conjecture, from the pioneering Coase (1972), held that in the case of monopolies providing durable consumer goods, assuming the value of the consumer goods does not decline (due to repairs, for example), the company's own products will end up in competition with ones bought at a different point in time, making it impossible to maintain monopolistic pricing. Taking a hint from this notion, a series of studies emerged on the topic of planned obsolescence and material obsolescence in particular (Bulow, 1986; Iizuka, 2007; Kuppelwieser, Klaus, Manthiou, & Boujena, 2019; Swan, 1972; Waldman, 1993; Wieser, 2016). In concrete terms, material obsolescence functions in the following manner:

- Making items difficult to repair (by raising the cost of repair, requiring special tools, etc.)
- Failing to provide information (for instance, manuals are not provided)
- Systematic obsolescence (making parts among models incompatible or making it impossible to fix newer models with parts from the older models)
- Numbering (frequently changing the model numbers to make it psychologically less attractive to use old models)
- Legal approaches (prohibiting access and modification to the internal structure of products by means of copyrights and patents)

Material obsolescence employs these kinds of approaches as a marketing technique for encouraging the consumer to replace an old model with a new one.³ Some also believe that as an industrial policy,

³ Economics has traditionally debated obsolescence in the sense of products deteriorating or becomes unusable, but its primary concern can be seen as obsolescence in the marketing context, i.e., how to stimulate demand for the replacement of goods which are still usable.

planned obsolescence is a requisite for technological progress, as encouraging rapid innovation through early obsolescence is better for the society than having products that last too long and hold back innovation (Fishman, Gandal, & Shy, 1993).

Planned obsolescence, which encompasses material obsolescence, had a particularly large impact on the American IT industry of the 1980s and 1990s; the reason was the notion that in addition to the physical hardware, which is durable in nature, to recover investments in software, which in principle does not deteriorate, assertions of legal rights and planned obsolescence is indispensable. The 1980s also saw the beginning of a pro-patent and pro-copyright wave as an industrial policy in the U.S., which was being battered by the technological competition with Japan, reinforcing the idea of protecting technology by legal means. It was a distinguishing characteristic of this era that the American justice system limited the application of anti-monopoly statutes, such as not deeming technology consortia to be cartels (Gibson & Rogers, 1994).

The Felten Case and the Right to Tinker

The Felten case

Planned obsolescence and the legal protection of software had the side effect of making it more difficult to research and develop software. This aspect gained widespread recognition through the Felten case about electronic watermarking.

Electronic watermarking is a technique for adding data to digital content such as music files in a manner imperceptible to human beings. The music industry at the time feared that digital music data ripped from CD media, for which there is no meaningful copy protection, would find its way onto the Internet, thus resulting in the proliferation of illegally copied music data. They believed that MP3

technology that enabled high levels of compression of music data with an attendant reduction in quality not noticeable to the human ear, and the widespread availability of portable players supporting MP3, would promote illegal copying through the Internet, thereby resulting in plummeting CD sales.

Digital watermarks, if the technology could be put in place, would have been able to be burned into the music data recorded on CDs, enabling devices to refuse to rip or play data if the watermark was not detected. Normally, data compression such as the MP3 format changes the data itself, losing any digital watermark. This aspect meant that if players based on the watermarking standard became common enough, although nothing could be done about existing CDs, the problem of illegal copying could be solved for newly published works. Conversely, if the music could be stripped of electronic watermarks, then new works could be played or ripped on old players, thus rendering copyright protection meaningless. Along these lines, in August 1999, Secure Digital Media Initiative (SDMI), using a technology developed by U.S.-based ARIS Technologies, published its “Open Letter to the Digital Community” (Craver, 2001). Simultaneously with the letter, SDMI published samples of the digital watermarking technology it had adopted—a particular music file, the same file with embedded digital watermark, and another music file with an embedded digital watermark—and issued a “challenge” to analyze the first two music files and strip the third file of its digital watermark with no severe degradation of audio quality; they set a duration of three weeks and a prize of US\$10,000.

Professor Ed Felten and his team from Princeton University took up the challenge and succeeded in removing the digital watermark. When Felten, who declined to accept the prize money, the condition for which was signing a non-disclosure agreement, attempted to present a paper giving the results of this analysis at an academic conference in 2001, he was issued a warning under the

U.S. Digital Millennium Copyright Act (DCMA) by SDMI, the Recording Industry Association of America (RIAA), and Verance (now Cinavia), a firm specializing in digital watermark technology for the music industry. The digital watermark technology that Felten had cracked was already in use in the market as a product made by Verance, who claimed that making public the means to remove it infringed DMCA's "anti-circumvention of technology protection measures," which we cover below.

Felten cancelled his presentation, but with the assistance of the Electric Frontier Foundation, then filed a suit to confirm that there was no legal issue with him publishing his paper. This suit was struck down due to lack of standing, but the U.S. Justice Department then guaranteed Felten and his researchers that the publication of their research had not been inhibited by the DMCA and that the suit against them based on DMCA was null and void, thus allowing Felten to present his paper, including the series of events that led up to it, at USENIX, another academic gathering.

The Felten case, while a suit fought over the allowability of publishing academic research, in essence questions the viability of the limitations the statute known as DMCA places on the Right to Tinker, which is necessary to be free to play digital data on one's portable music player. Many similar cases were to follow. In that sense, the Felten case was the model case during the first stage of questioning the loss of the Right to Tinker imposed by technical and statutory means.

The Right to Tinker

Samuelson (2016) brought together the following eight aspects of the freedom to tinker based on the Felten case.

1. Intellectual freedom to study existing artifacts in depth and conceive what can be done with them.

2. Freedom to intellectual privacy and autonomy to investigate and experiment with artifacts one owns or has a legitimate interest in within the confines of one's own space.
3. Freedom to improve one's skills by testing, analyzing, and interoperating existing artifacts.
4. Freedom of individual self-expression by means of tinkering.
5. Freedom to learn from tinkering and convey the research results to others.
6. The right to fix broken things and the freedom to use artifacts for purposes other than originally envisioned to the extent that this aspect does not impinge on others' rights.
7. Freedom to innovate on the basis of the learnings from tinkering.
8. Freedom to optionally share innovations arising as a result of tinkering with others and build communities with such innovations at their core.

The abovementioned factors express the freedom to tinker as the right to do as one wishes with things one owns; this aspect is nearly identical to the textbook definition of modern ownership rights, i.e., the right of complete control over things and to use them, make profits from them, and dispose them.

The Return of “the Right to Repair”

The 2000s brought slightly more momentum toward legislating the Right to Repair. This aspect was brought about by the wide use of smartphones and particularly by the issuance of Apple's policy of allowing their iPhone products to be repaired only at the company-owned Genius Bars, prohibiting disassembly and repair at third-party shops under the DMCA. In response, a group named The Repair Association was formed in 2013, representing repair shops and consumers, and kicked off a movement to legislate the Right to

Repair in South Dakota, New York, Minnesota, and Massachusetts. The movement succeeded in California in 2019, and is expanding in the European Union (Gault & Koebler, 2020).⁴

The rising visibility of the Right to Repair in recent years is connected with the advances in innovation research. Torrance and von Hippel (2013) considered the legal regulations and the nature of intellectual rights in particular as factors either nurturing or impeding innovation and proposed the concept of “innovation wetlands.” They relate how, in the U.S., wetlands were traditionally considered “noxious sources of pestilence and disease” and “malarial swamps,” with 550,000 acres of wetlands being landfilled annually between the 1950s and 1970s. Advances in the biological research commencing in the 1950s revealed that far from being worthless areas, wetlands were the most productive regions in the earth’s ecosystem, playing an extremely important role in biological diversity, controlling floods, and purifying water. Based on these insights, the U.S. Clean Water Act of 1972 was amended in 1972, mandating legal protection for wetlands. In 1975, the so-called Ramsar Convention came into effect, thus calling for the protection and revival of wetlands internationally.

Torrance and von Hippel (2013) dubbed the set of circumstances in which the rights and conditions necessary for brisk individual innovation are preserved as the “innovation wetlands.” Their focus was the so-called user innovation. The individuals who perform user innovation possess only modest financial and other resources and expect only modest rewards. There is a serious danger that more stringent regulations by governments or others will have a major chilling effect on such actors compared to large enterprises with ample resources. Torrance and von Hippel (2013) argued that just as diverse organisms coexist in a delicate balance in the fragile

⁴ See Table 1, [3][4][6]

ecosystem of actual wetlands, and thus require aggressive protection, innovation wetlands, as an ecosystem fostering user innovation, must also be explicitly protected as a matter of policy. Their mention of the necessity of legally protecting those spaces where innovation is born is very much in the spirit of the second of the eight aspects of Samuelson's (2016) freedom to tinker.

Thierer (2016) proposed another concept, i.e., "permissionless innovation." He argued that previous policies to promote innovation were wrapped up in individual policies and that the focus should be on social values underpinning innovation, such as cultural attitudes, social norms, and political viewpoints. From this perspective, he makes a broad distinction between permissionless innovation and the precautionary principle in the social value system.

Permissionless innovation is the notion that new technologies and experimental business models should as a general principle be permitted, unless it can be proven that they clearly cause serious damage. The precautionary principle, in contrast, is the idea of preventing new innovation until it can be proven that that innovators will cause no damage whatsoever to individuals and groups, specific actors, cultural norms, various existing laws, or tradition. According to Thierer (2016), the vitality of innovation and dynamism in a society will be governed by which of these two notions is dominant in a particular society.

The concept of permissionless innovation is suggestive in considering how society fundamentally assesses the Right to Tinker. In particular, given the strong emphasis on safety and security in Japan, we believe that it is necessary to actively protect the Right to Tinker as a policy to achieve a balance.

Conclusion

This paper has examined the history, specifics, and significance of

the Right to Tinker and the Right to Repair, which although in existence for many years and have been serving as the precondition for innovation, were ignored simply because they were taken for granted. We thus highlighted how the Right to Repair, at one point in retreat, has enjoyed resurgence as the importance of the Right to Tinker has been reassessed.

While the existence of the Right to Tinker and the Right to Repair, which lie at the heart of innovation, have been threatened in recent years in the form of legal restrictions, advances in technology have placed more power in the hands of the ordinary users, thus giving the Right to Tinker, originally not extending beyond the use of one's devices, a broader influence in the world. Further, to strike this balance will be important in the future discussions related to these rights. We also hope to address these issues in detail as we continue our research.

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